

ITEC 300: Project 1: R Package Investigation & Demonstration

Objective: The purpose of this assignment is to help students explore the diverse ecosystem of R packages available for data analytics, gain experience in independently researching tools, and develop presentation skills to effectively communicate technical content.

Instructions:

1. Package Selection Requirements:

- Each student must choose a unique R package that will not be covered in class¹. This could include packages for data manipulation, visualization, machine learning, statistical analysis, or any other area of interest.
- The package must contain significant functional capabilities. This means, the package must be more than just a dataset. A dataset (like diamonds in ggplot2) can be a part of the package, but it cannot be the only element of the package.
- Package selection will come on a first-come first-serve basis and will be confirmed with instructor approval. You will sign up for your R package in our Canvas discussion board. This will ensure that no two students present the same package.

2. Deliverable 1: Package Whitepaper - Research and Demo Creation:

- You will use RMarkdown to create a white paper for the R package you select. Content should include:
 1. **Purpose:** Explain what the package is (primary function). Explain where the package would appropriately be used (applications/ typical use cases). Discuss why the package is useful (*for this you might want to think about the alternatives*).
 2. **Core Functions w/ Example Analysis:** Highlight at least **three** key functions or features of the package. Provide examples of how these functions are used in a real-world scenario. Find a dataset to highlight your package functionality. Create a simple, reproducible example that showcases the package in action. This could be a brief analysis, a visualization, or any other relevant demonstration, this will be largely dependent on the package you select. **NOTE:** *these examples cannot be copied from existing package documentation, you must create your own.*
 3. **Advantages and Limitations:** Discuss the strengths and any potential drawbacks or limitations of the package. Make sure to discuss which of the Three Core Principles of R² this package relates to the most and why.

3. Deliverable 2: In Class Presentation

- You will have three minutes to showcase the R package you've researched. The presentation should be three minutes long and cover the topics included in deliverable 1. To showcase the R package, you must have either slides, or a live

coding demonstration to highlight the core capabilities. Students will be randomly selected to present between the two presentation dates.

- For those using slides, ensure the visuals are clear, concise, and support the narrative of your R package. For those opting to do a live demo, ensure you are prepared and have a clear environment free of irrelevant or distracting files.
- Audience: deliver the content in a manner that is easy to understand even for an audience who is unfamiliar with the package.

¹You cannot select a base R package, or one of the following packages: knitr, markdown, rmarkdown, ggplot2, ggthemes, esquisse, lubridate, openxlsx, chron, ctv, tseries, rvest, foreign, MASS, plyr, dplyr, ISLR, GGally, Hmisc, corrplot, rpart.plot, tidytext, schrute, psych, tree, car, shiny, shinydashboard, shinydashboardPlus. You may select a tidyverse package as long as it's not explicitly listed above.

²If you choose reproducibility, you must have a deeper explanation other than it's a package or function that allows you to replicate content. That claim can be made about every package – what makes this one different is what we want to highlight.